3DreamBooth: High-Fidelity 3D Subject-Driven Video Generation Model

Hyun-kyu Ko, Jihyeon Park, Younghyun Kim, Dongheok Park, Eunbyung Park | Yonsei University

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Research Questions

- How can video generation models generate view-consistent videos of customized 3D subjects without requiring large multi-view video datasets?
- How can spatial geometry be decoupled from temporal motion in video diffusion models to prevent temporal overfitting during subject customization?
- How can multi-view conditioning be efficiently integrated into video diffusion models to preserve 3D identity while maintaining computational efficiency?

Core Idea

Existing subject-driven video generation methods treat subjects as 2D entities, failing to maintain 3D consistency across novel viewpoints. The key insight is that video diffusion models already possess implicit 3D priors from their training, which can be explicitly leveraged through a 1-frame optimization paradigm that naturally bypasses temporal attention mechanisms. The paper introduces 3DreamBooth (which bakes 3D identity into a unique token via multi-view single-frame training) combined with 3Dapter (a multi-view conditioning module that acts as a dynamic selective router). This approach decouples spatial geometry learning from temporal dynamics, preventing overfitting while enabling view-consistent video generation.

Key Formula

3DreamBooth Training Objective

$$\mathrm{argmin}_{\phi_{3DB}} E_{i,\epsilon,t}\left[\|\bar{v} - v_{\theta,\phi_{3DB}}(z_t^{(i)}, t, p)\|_2^2\right]$$

i is sampled view index, z_t^{(i)} is noisy latent of view s^{(i)} at timestep t, \bar{v} is target velocity vector, p is text prompt, \phi_{3DB} are LoRA weights. This loss trains on single frames (T=1) to learn multi-view geometric variations while preserving temporal priors.

Joint Optimization Objective

$$\mathrm{argmin}_{\phi_{3DB},\phi_{3Dapter}} E_{i,\epsilon,t}\left[\|\bar{v} - v_{\theta,\phi_{3DB},\phi_{3Dapter}}(z_t^{(i)}, t, X, p)\|_2^2\right]$$

X represents conditioning views, \phi_{3Dapter} are 3Dapter weights. This jointly optimizes 3DreamBooth and 3Dapter parameters where 3Dapter provides multi-view conditioning while 3DreamBooth reconstructs target view, enabling efficient 3D-conditioned generation.

Key Figure

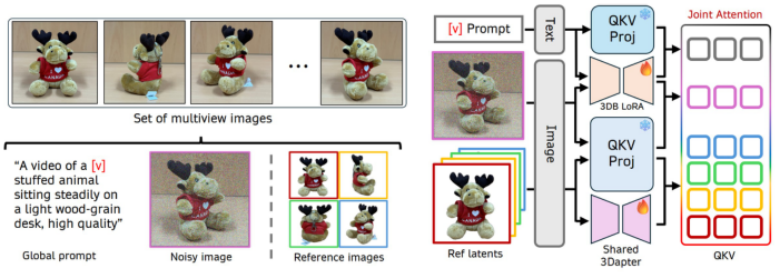

Fig. 2: Overview of the 3DreamBooth training pipeline. (Left) From multi-view images, one is selected as the target, while a sampled subset serves as reference conditions alongside a global prompt with a unique identifier V. (Right) The text and noisy target latents pass through the main branch (3DB LoRA), while reference latents pass through a shared *3Dapter*. Their features are concatenated for Multi-view Joint Attention. This 1-frame optimization decouples spatial geometry from temporal dynamics to efficiently learn a 3D prior.

successfully binds a unique identifier (e.g., a rare token V) to a specific subject by fine-tuning the model to reconstruct the subject's appearance. However, extending this concept directly to video generation models requires a deeper understanding of the interplay between spatial representation and temporal dynamics.

Isolating Identity from Temporal Dynamics. Typically, video diffusion models are trained on large-scale datasets to learn both appearance and motion. However, when the objective is to inject the identity of a specific subject, utilizing full video sequences of the subject for training is computationally redundant and highly prone to temporal overfitting (e.g., the model memorizing a specific motion trajectory). Based on the insight that object identity is largely a spatial attribute, we propose a 1-frame video training paradigm.

Modern video Diffusion Transformers (DiTs) [28] often process inputs via joint spatio-temporal attention. While recent video customization methods typically require explicitly freezing temporal modules [13] or inserting separate spatial adapters to learn subject identity from static images [44, 52], our approach leverages an inherent architectural property: when the input is restricted to a single frame ($T = 1$), the temporal attention mechanism is naturally bypassed. This effectively localizes all gradient updates exclusively to spatial representations without requiring explicit architectural modifications. Consequently, we uniquely harness this mechanism to implant the subject's comprehensive 3D visual identity into the model while implicitly preserving its pre-trained temporal priors. During inference, these untouched temporal mechanics naturally extract and drive the temporal flow of the learned identity, enabling smooth, view-consistent video generation.

Figure 2: Shows the complete 3DreamBooth training pipeline architecture, illustrating how multi-view images are processed through the dual-branch system with 3Dapter conditioning and Multi-view Joint Attention for 1-frame optimization.

Limitations

- Requires multi-view reference images of subjects, which may not always be available for arbitrary objects
- Limited to subjects that can be effectively captured from multiple viewpoints, potentially excluding complex or occluded objects
- Computational overhead from processing multiple reference views during inference, though reduced compared to training-based approaches
- Performance depends on quality and coverage of reference viewpoints - insufficient angular coverage may lead to artifacts in novel views